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SUBJECT NAME: Power BI And Tableau Practical

PRACTICAL - 9

AIM: Cleaning and Structuring Data for Analysis

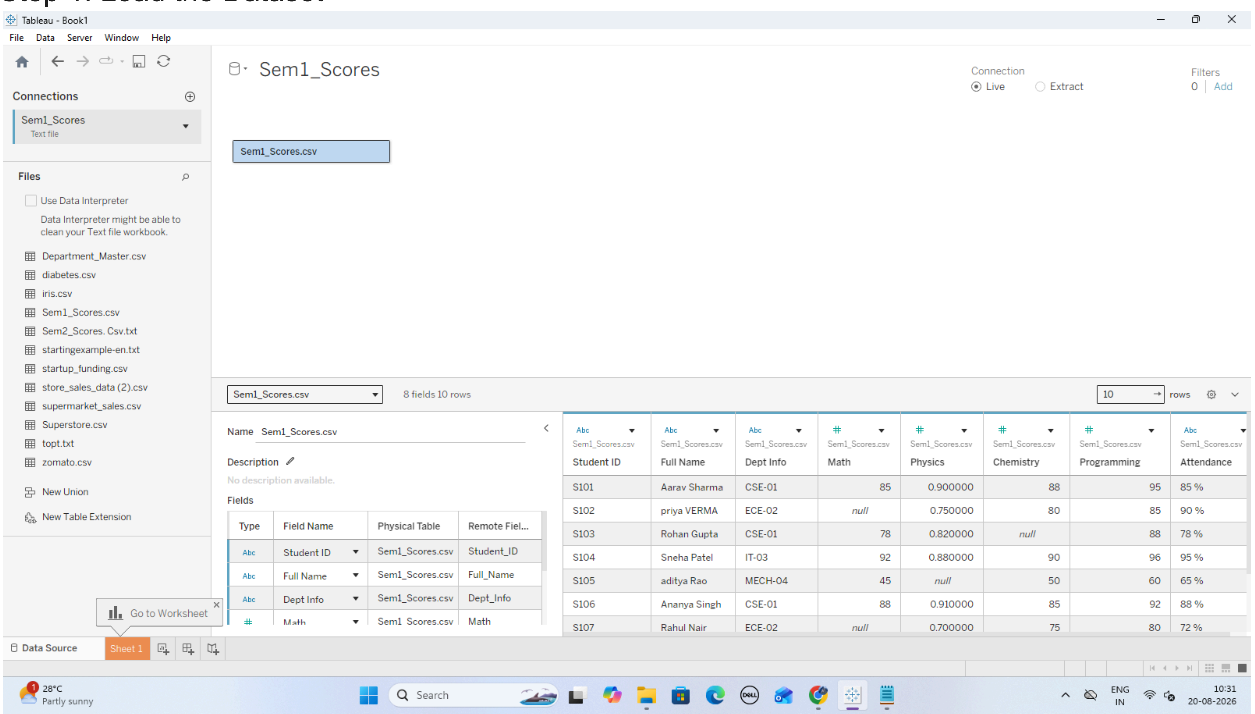
a) Identify and handle missing or inconsistent data
b) Convert wide data into tall format
c) Perform union of multiple files
d) Execute cross-database joins
e) Restructure data for better visualisation
f) Handle different levels of detail in datasets
g) Apply advanced data cleaning techniques

Dataset: Custom Designed Students Dataset [Sem1_Scores.csv, Sem2_Scores.csv, Department_Master.csv and Student_Activities.csv]

Steps:-

A: Identify and Handle Missing or Inconsistent Data

Step 1: Load the Dataset



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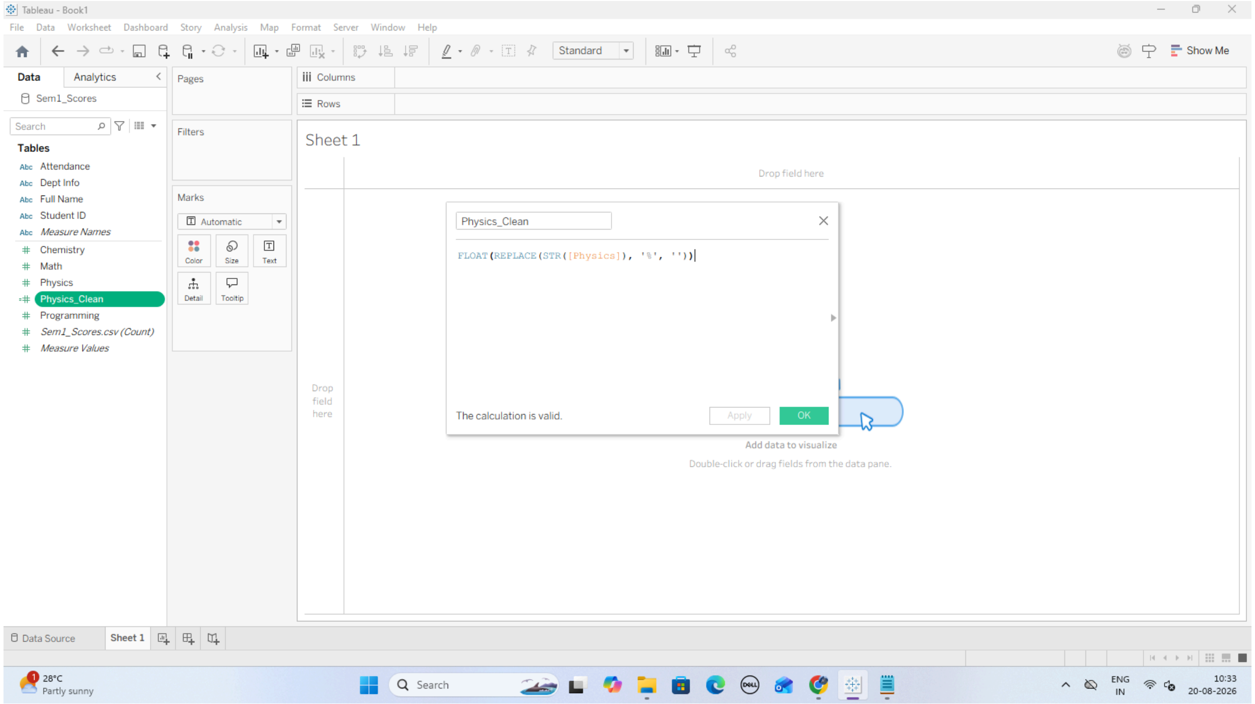
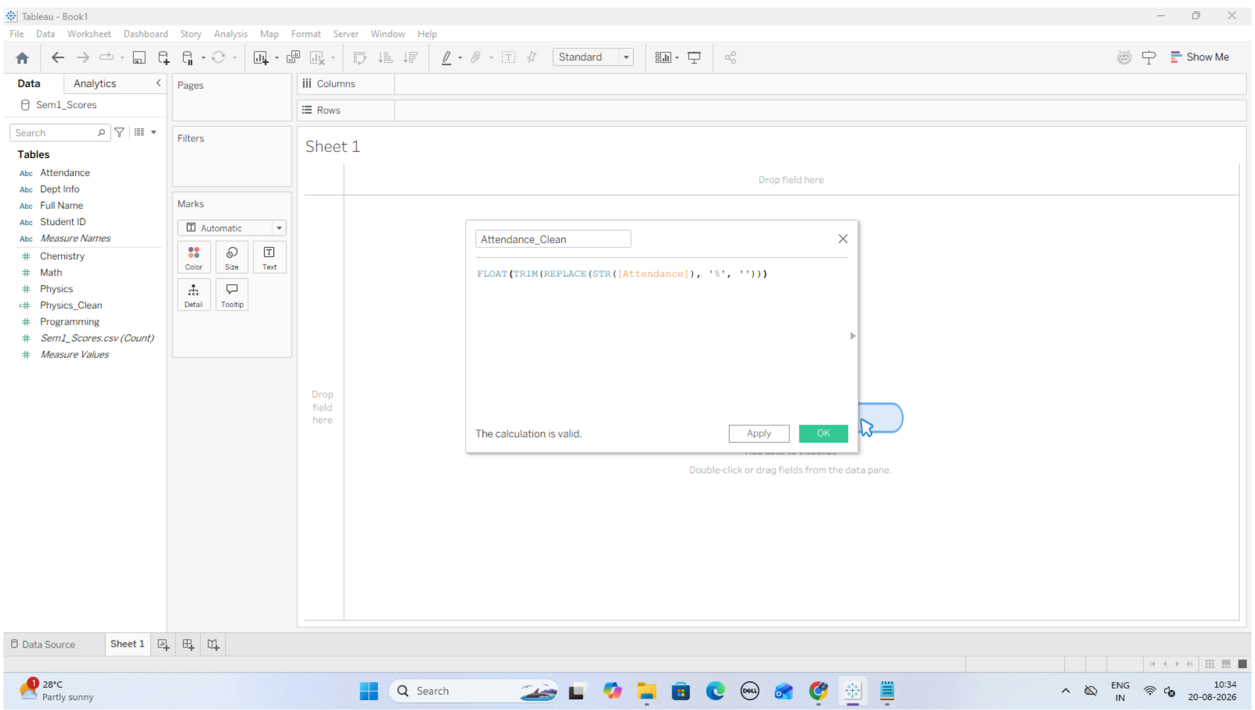
Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)





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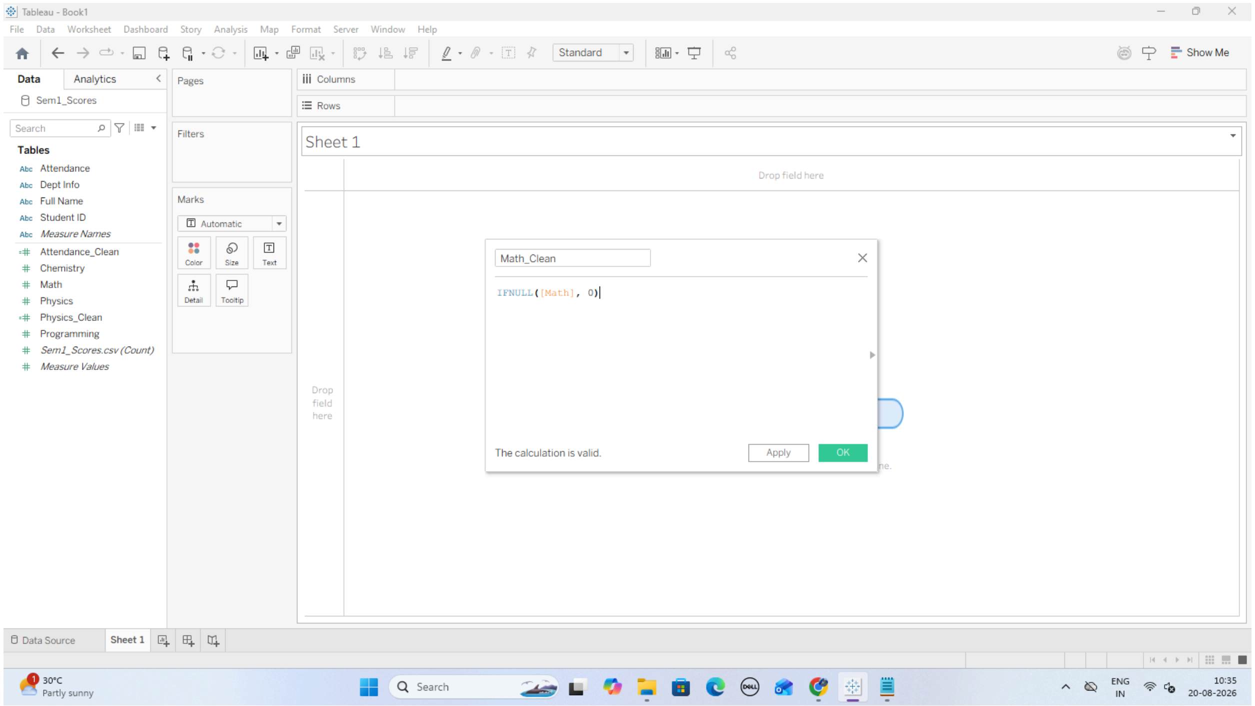
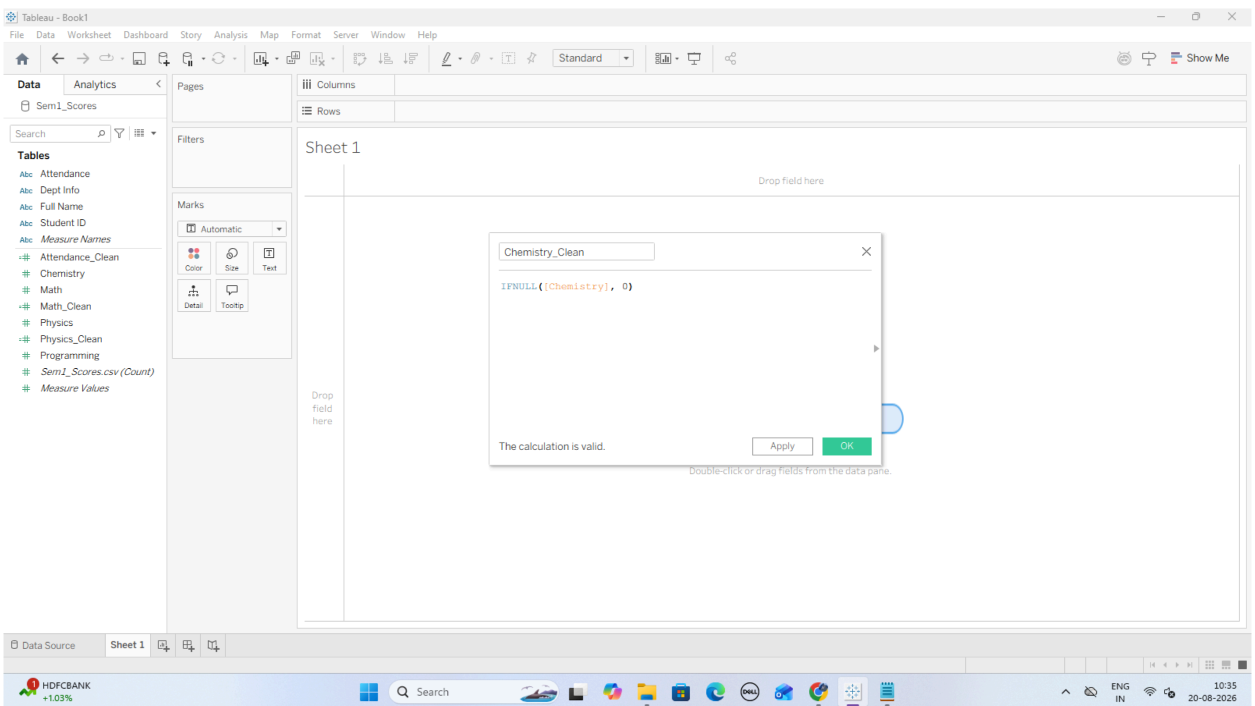
Step 3: Handle Missing (Null) Numeric Values via Imputation





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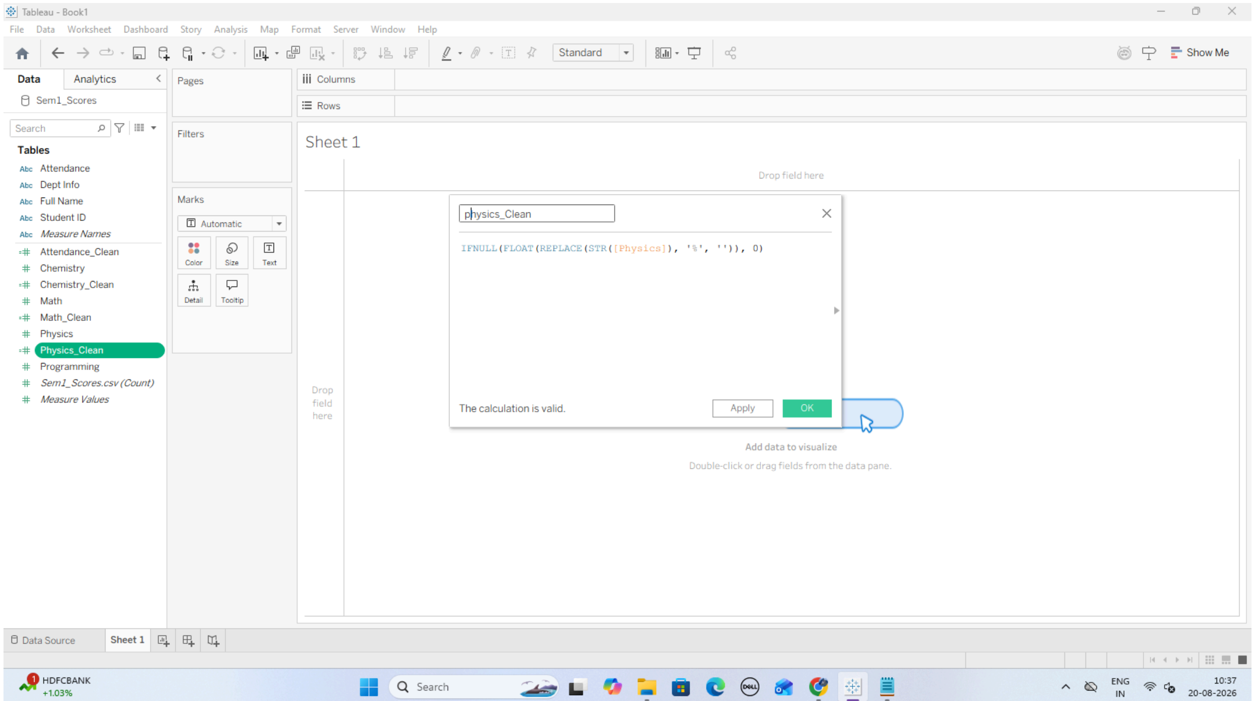
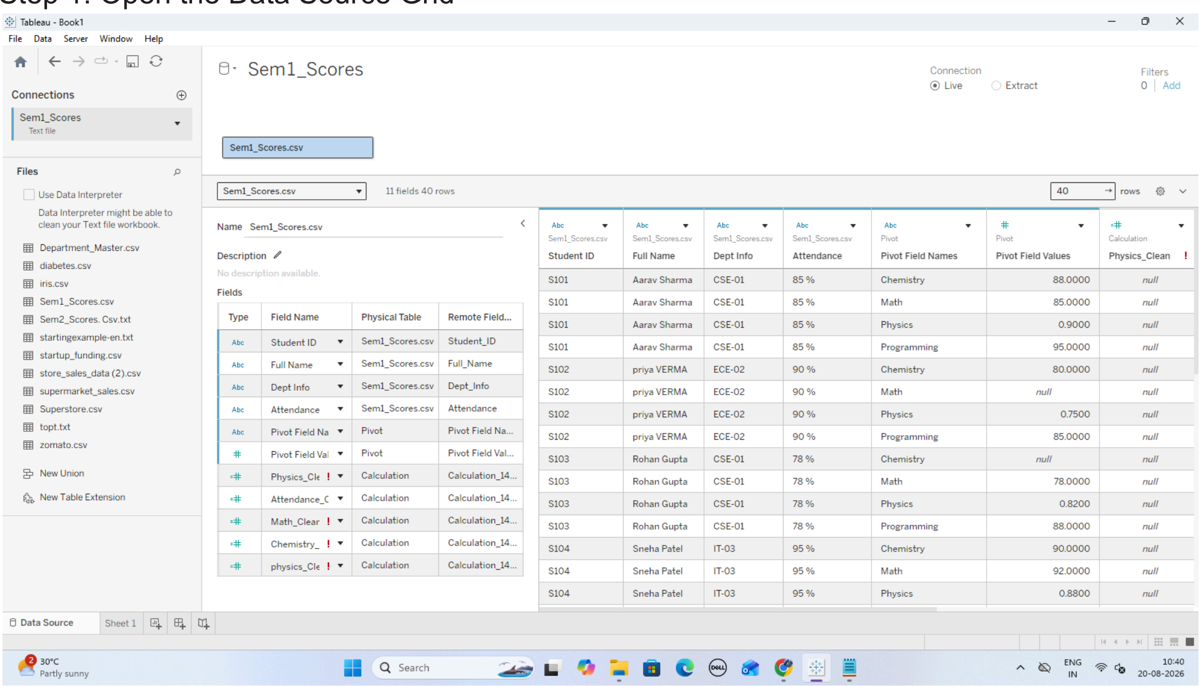
Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)



B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid



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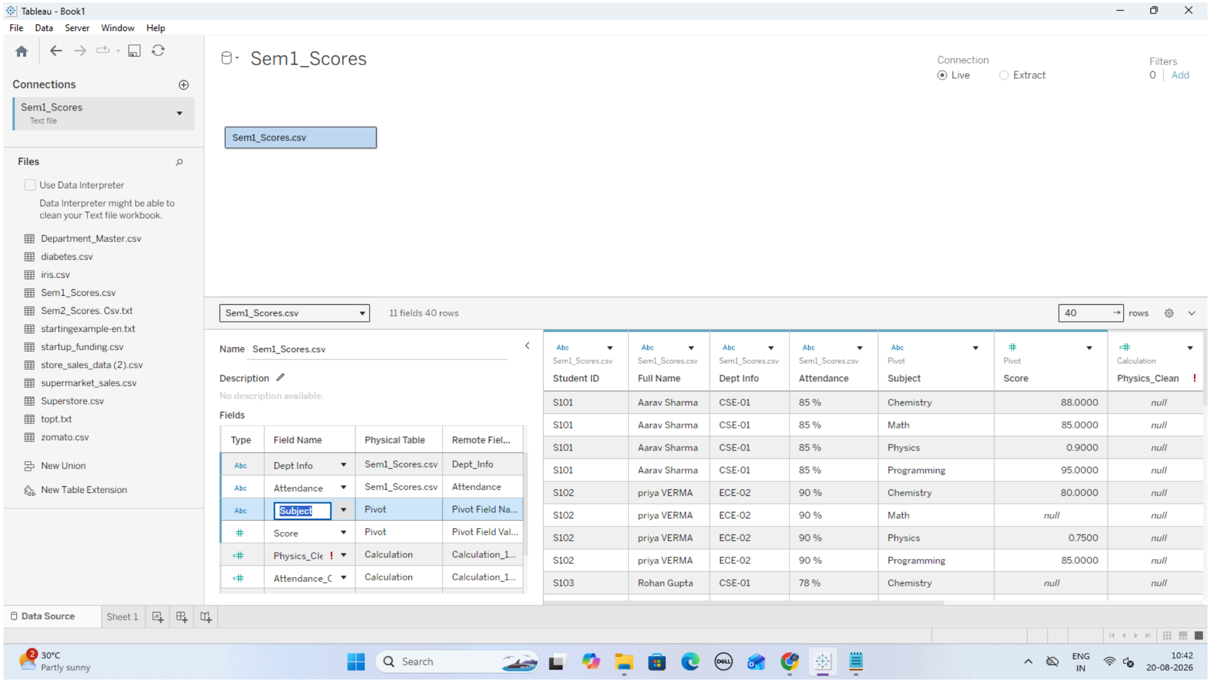
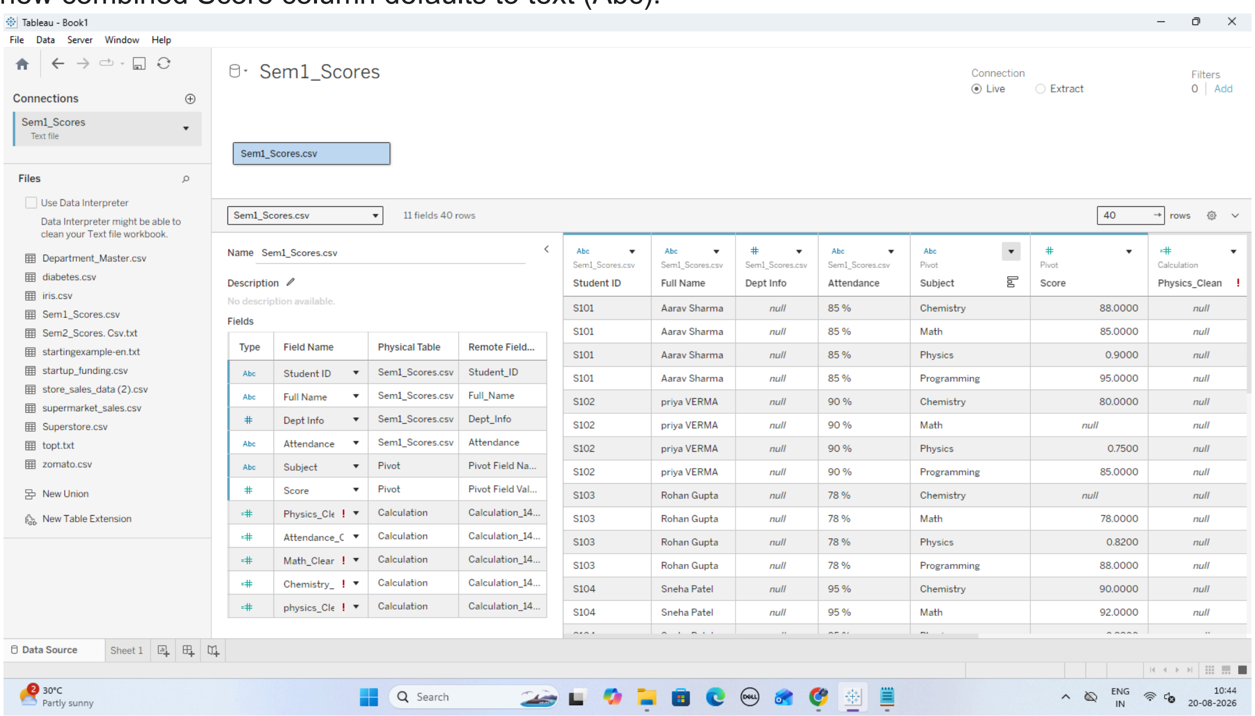
Step 2: Select multiple columns and choose **Pivot**; Tableau consolidates the selected columns into two fields, **Pivot Field Names** and **Pivot Field Values**, which can then be renamed as needed.



Step 3: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc).



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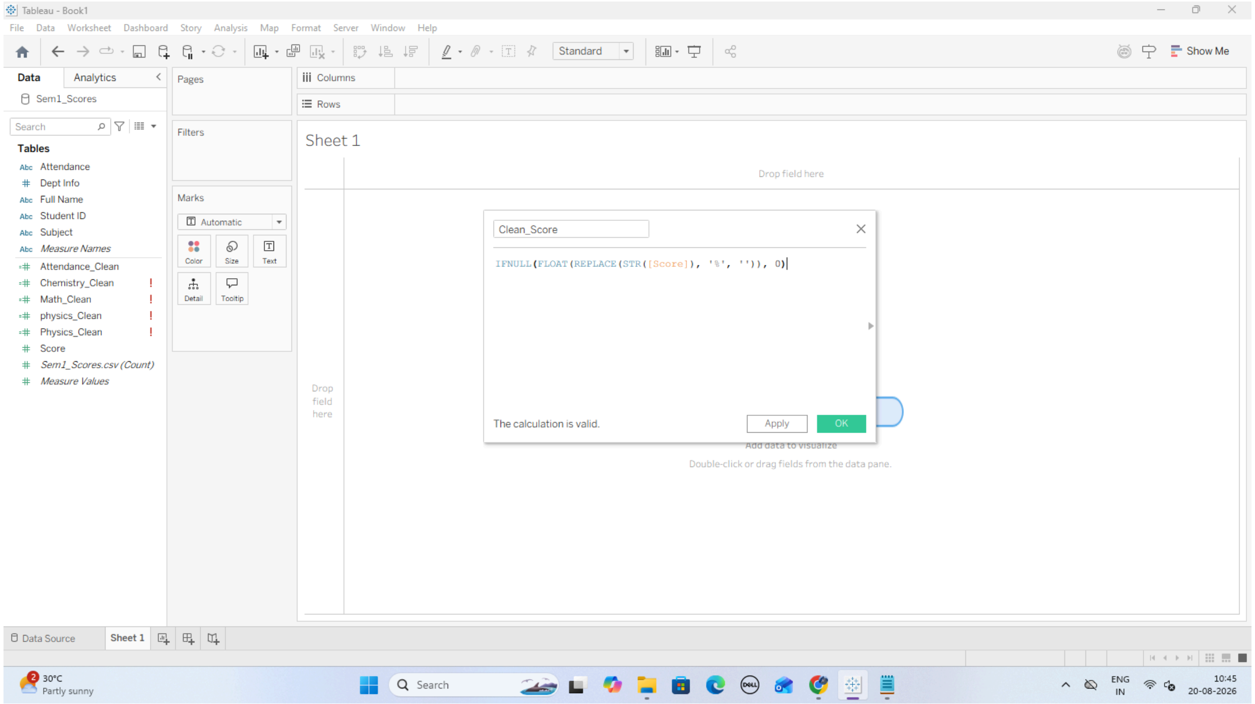
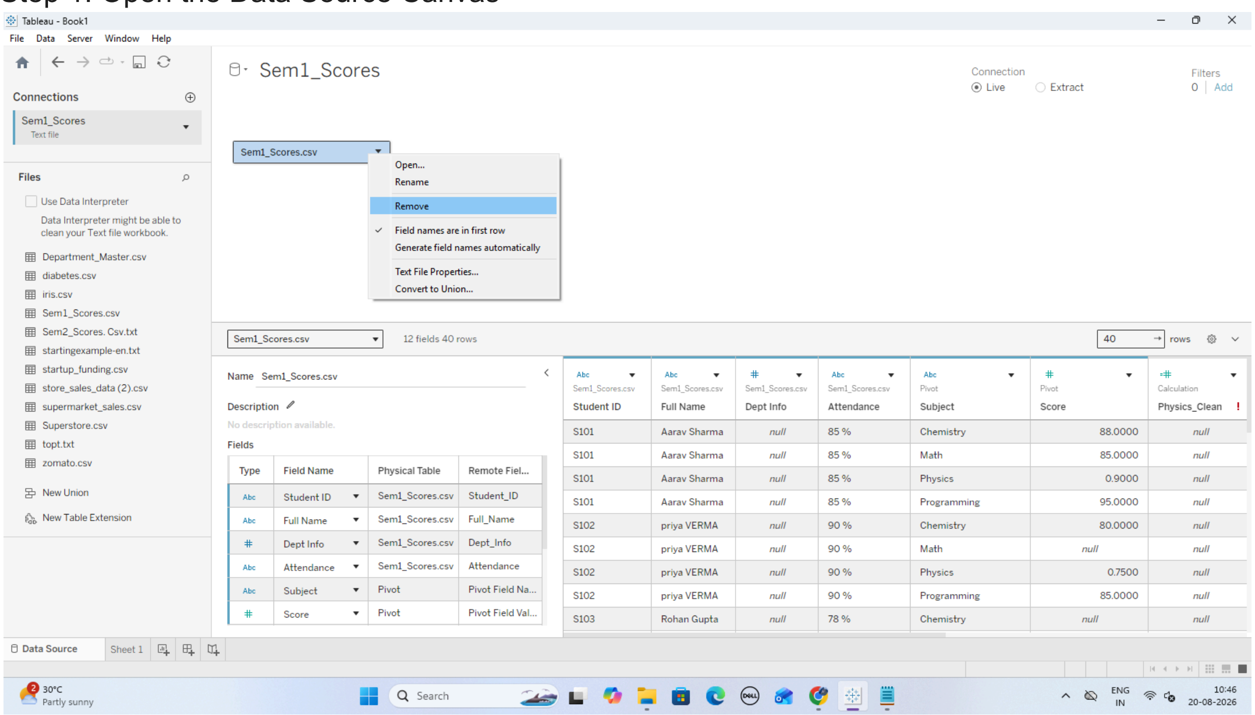


C: Perform Union of Multiple Files

Step 1: Open the Data Source Canvas



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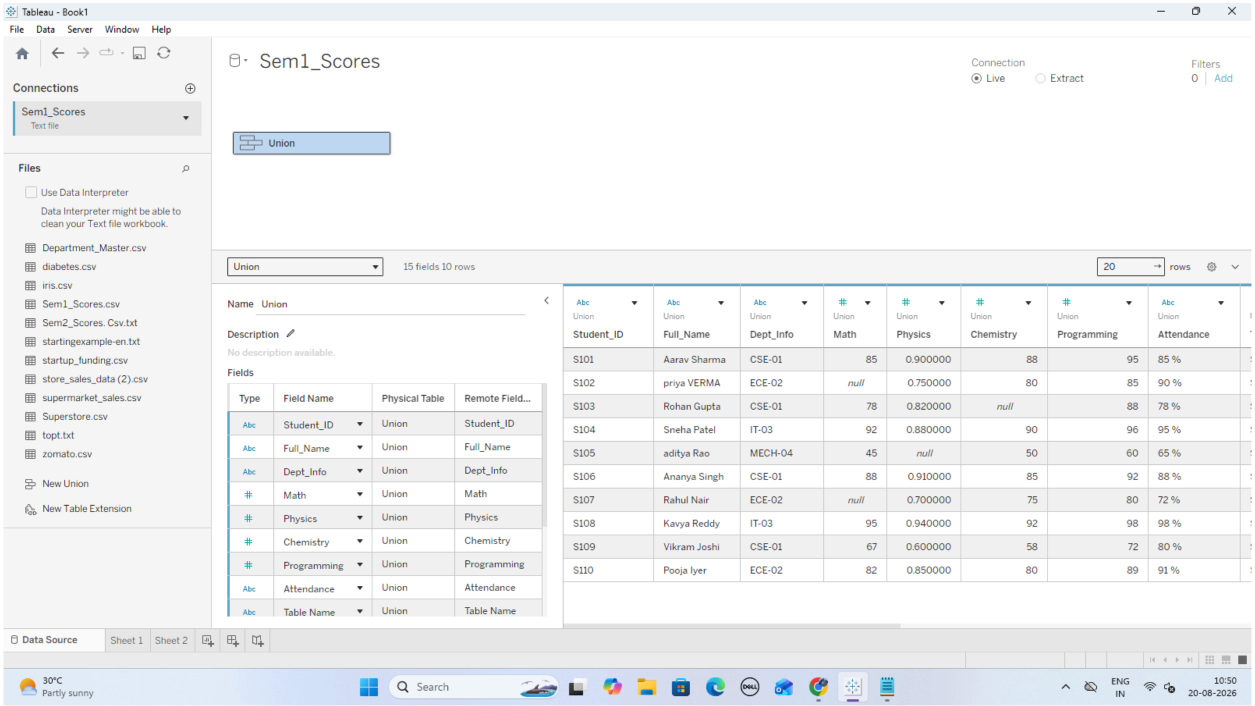
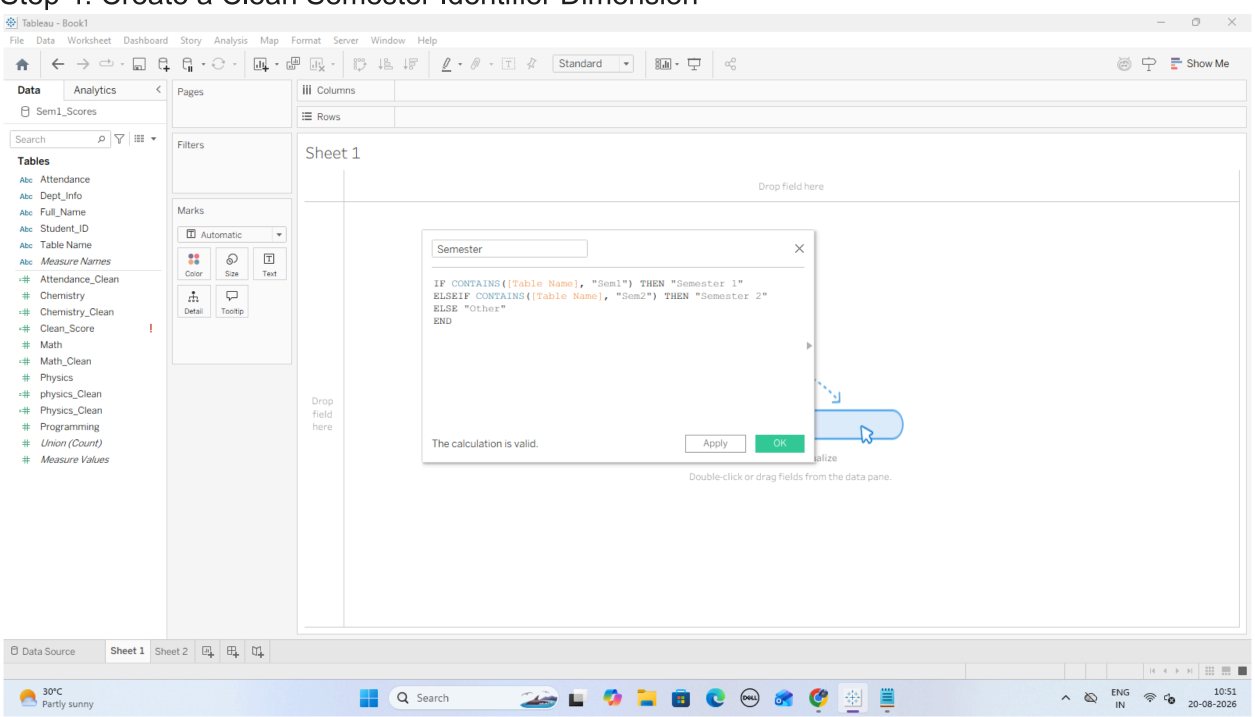
Step 2: Create a Multi-File Union



Step 3: Verify the Merged Data Grid

Step 4: Create a Clean Semester Identifier Dimension



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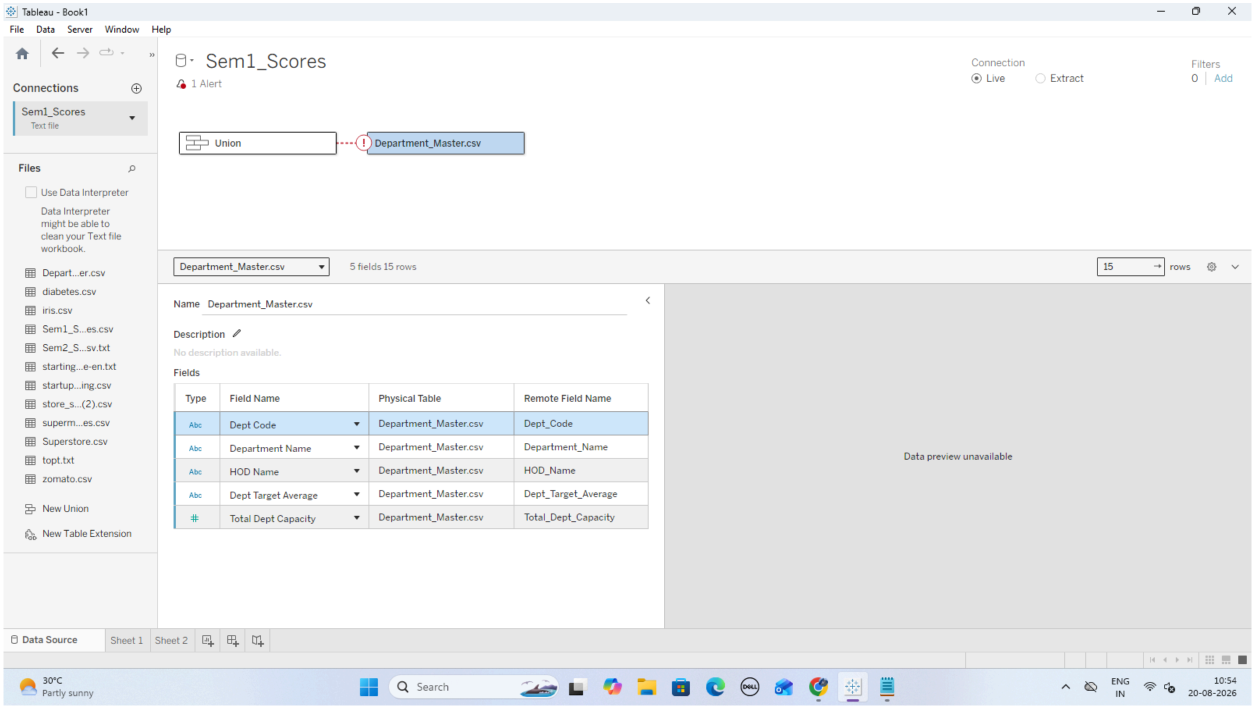
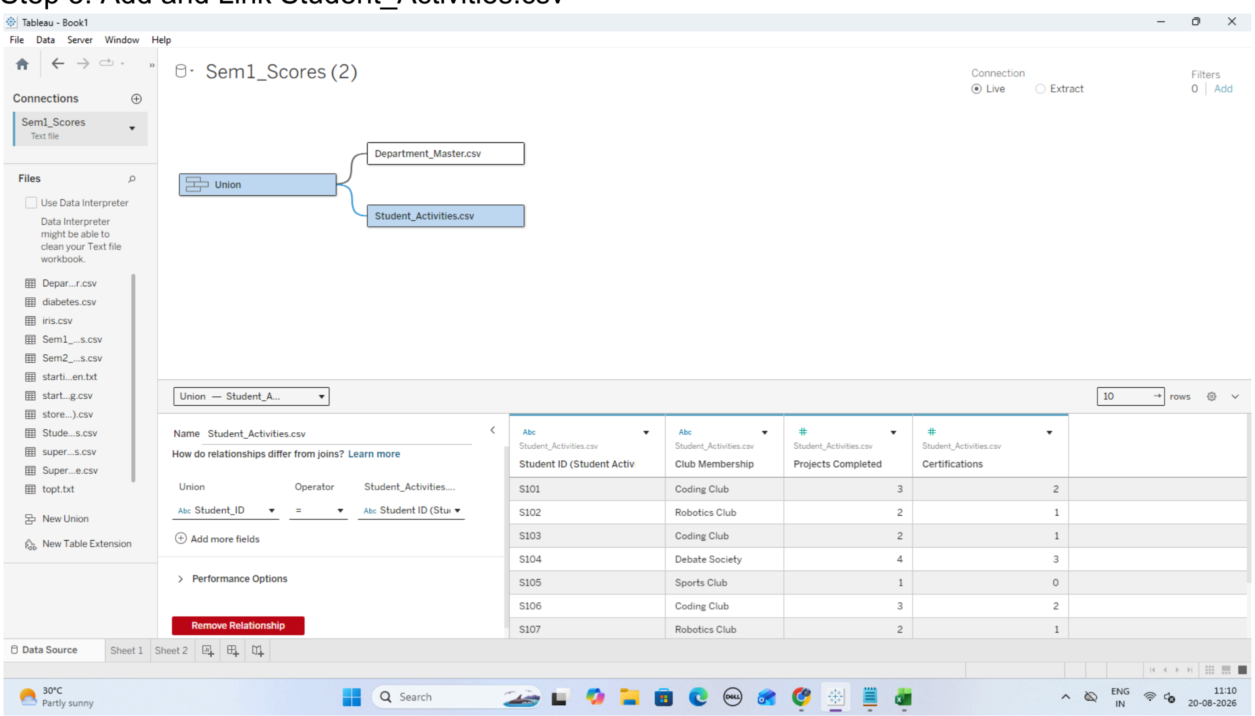
D: Execute Cross-Database Joins



Step 1: Stay on the Top Canvas
Step 2: Add and Link Department_Master.csv
Step 3: Add and Link Student_Activities.csv



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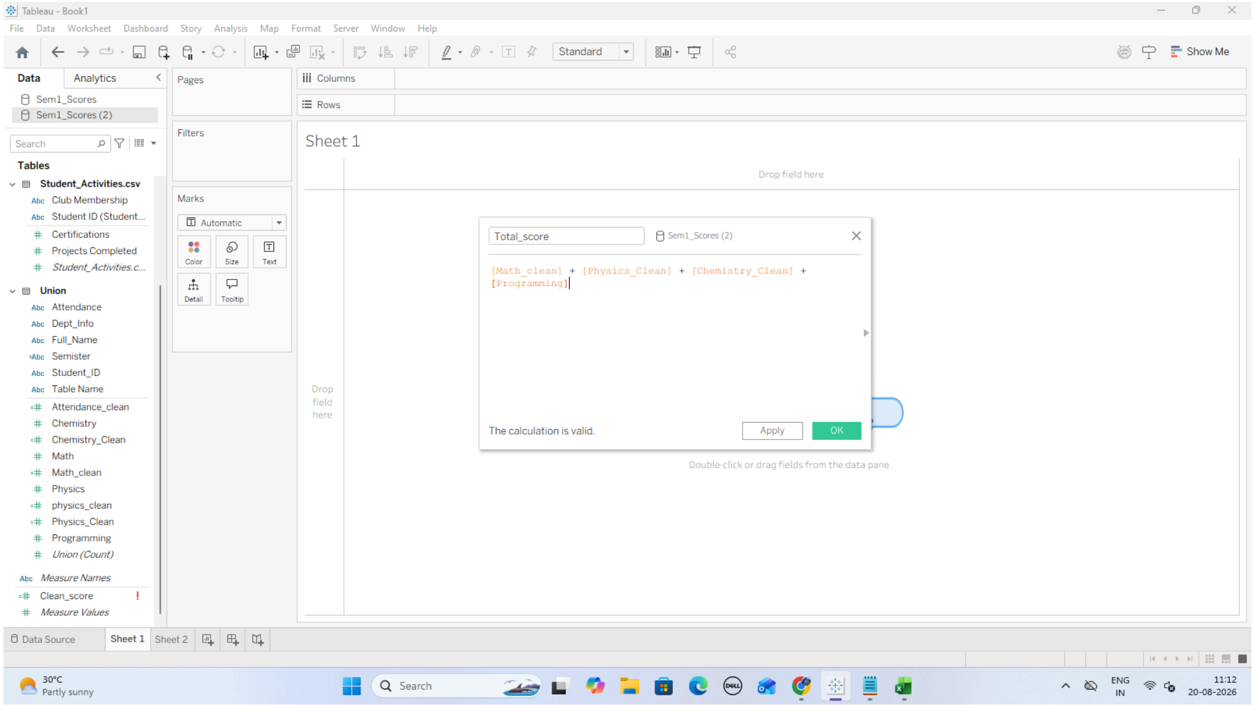
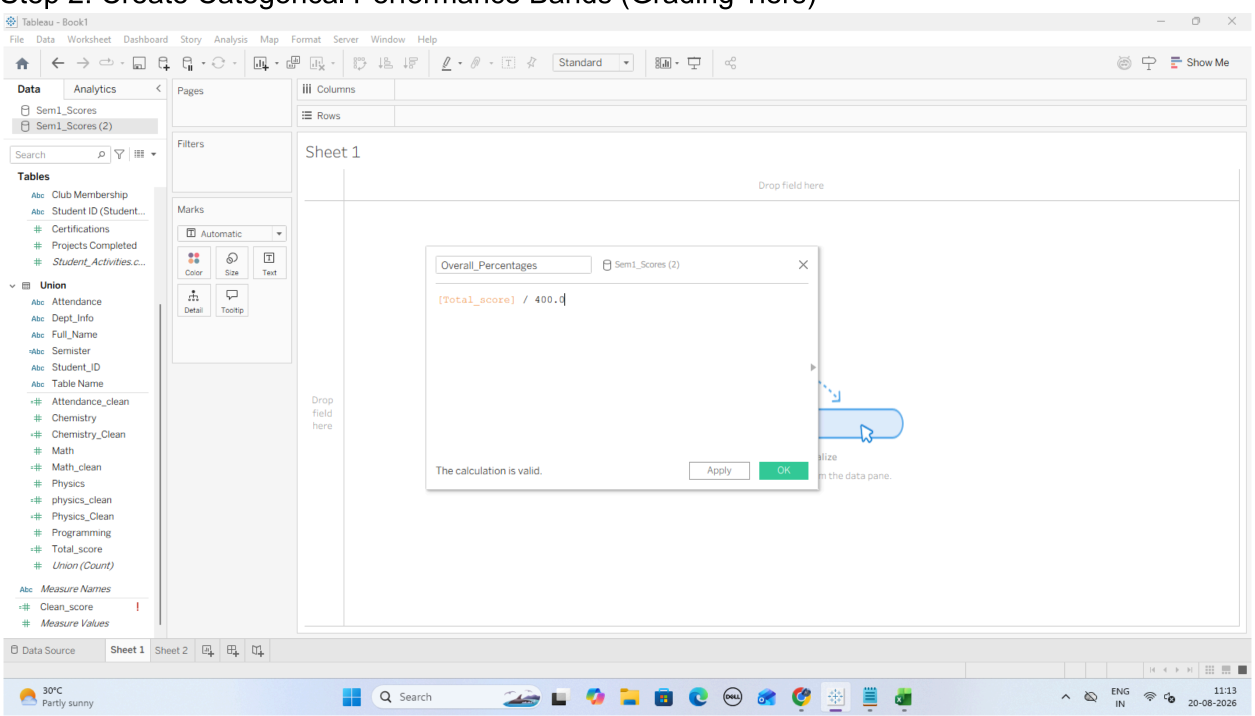
E: Restructure Data for Better Visualization

Step 1: Calculate Total Marks and Scaled Percentage



Step 2: Create Categorical Performance Bands (Grading Tiers)



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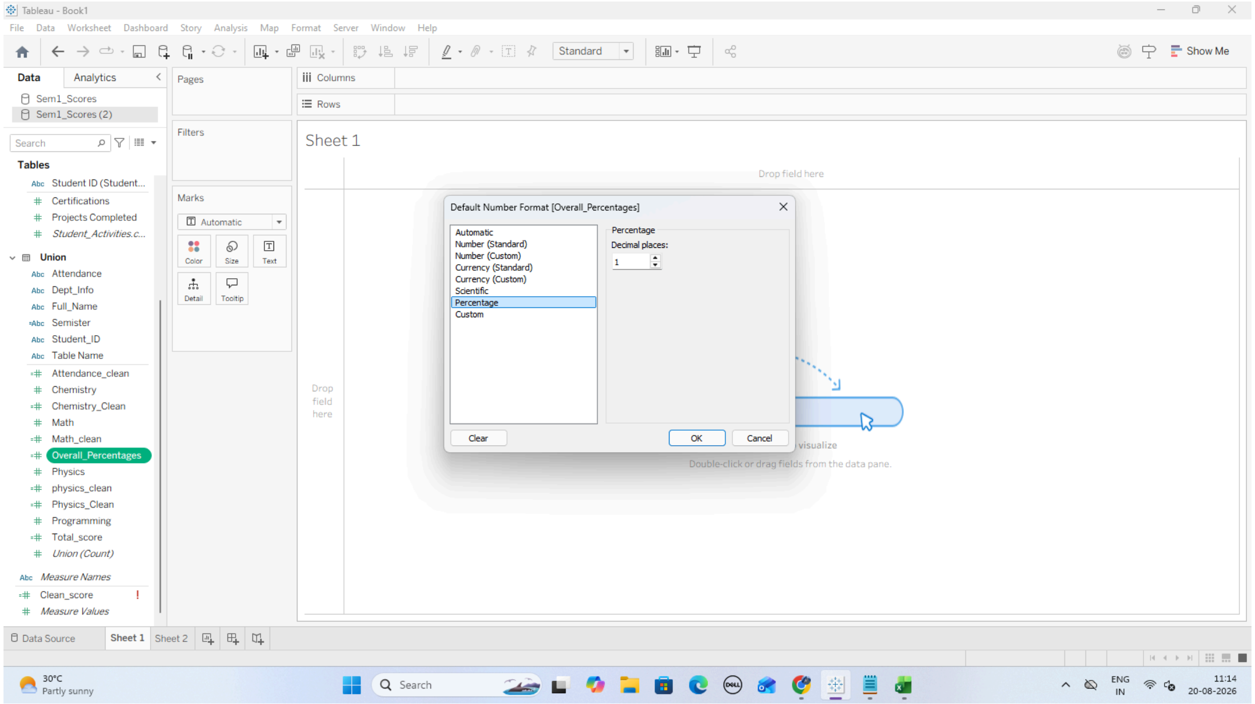

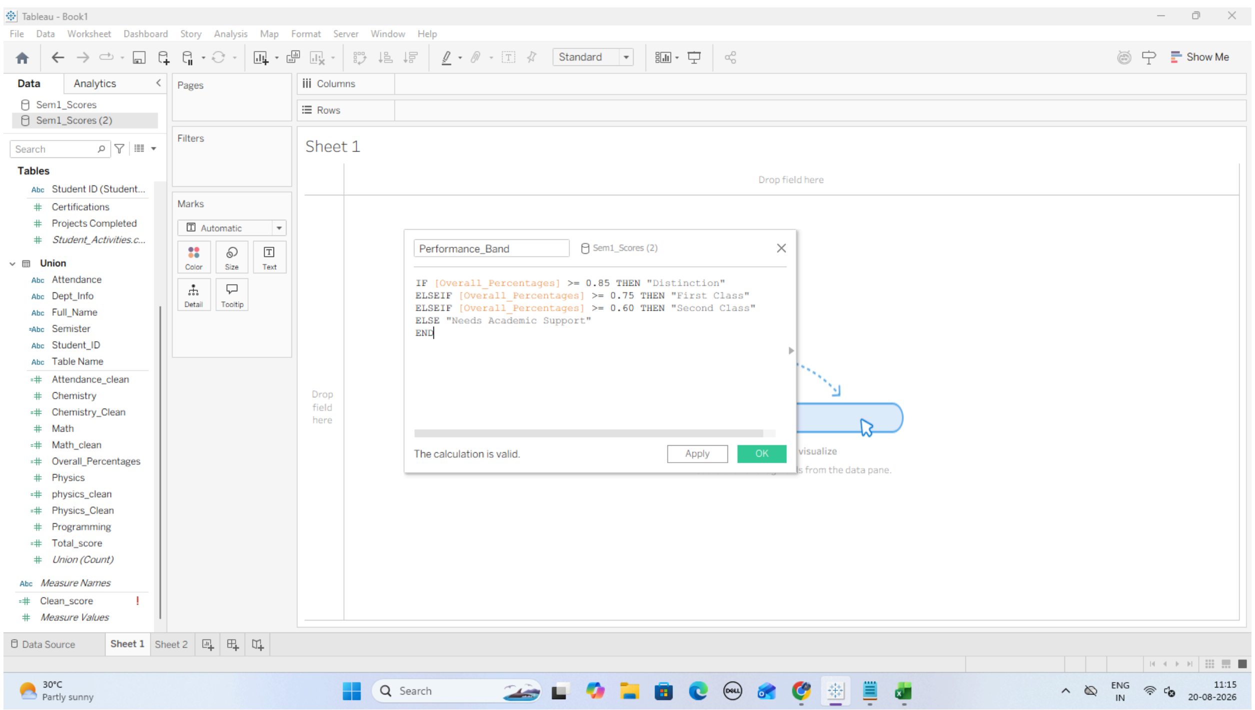

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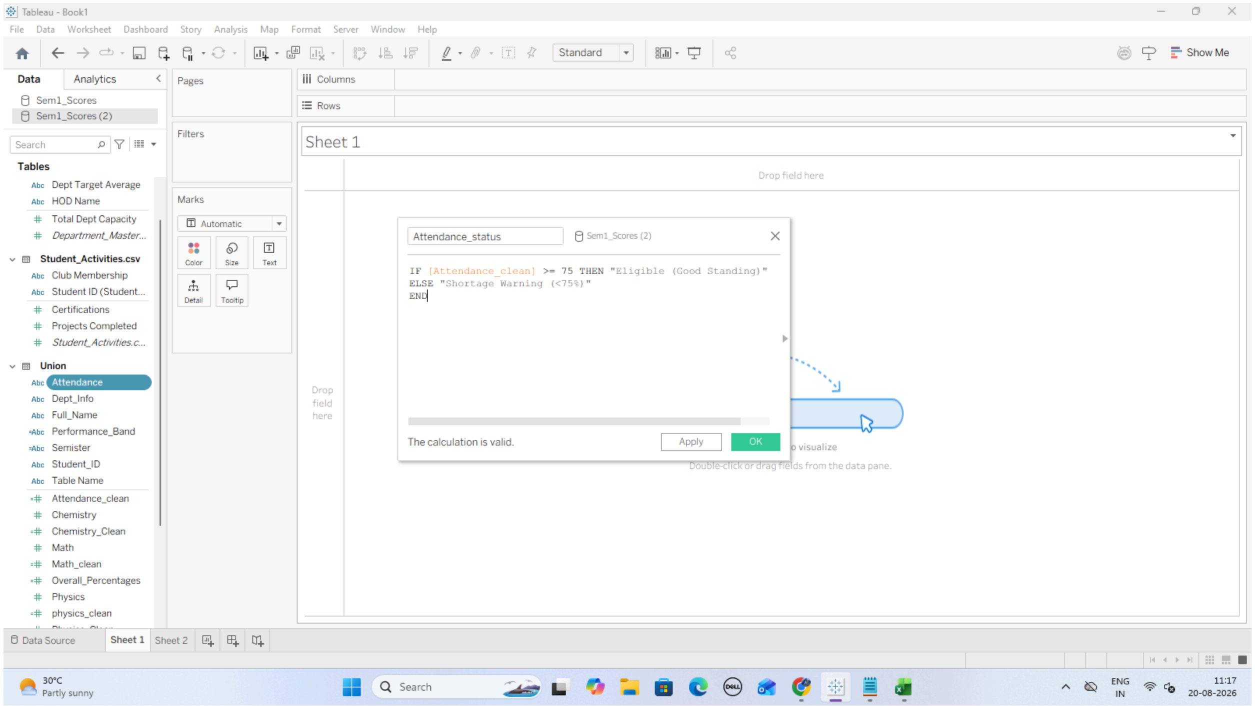
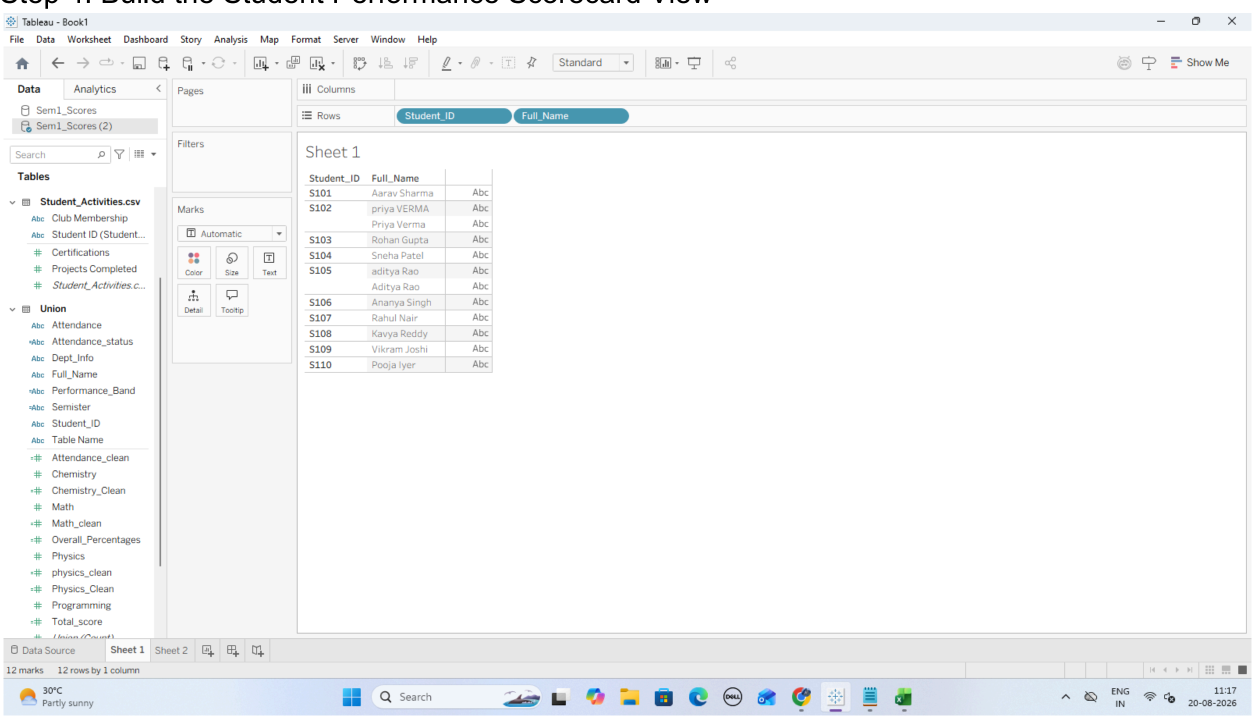
Step 3: Create an Attendance Health Flag



Step 4: Build the Student Performance Scorecard View



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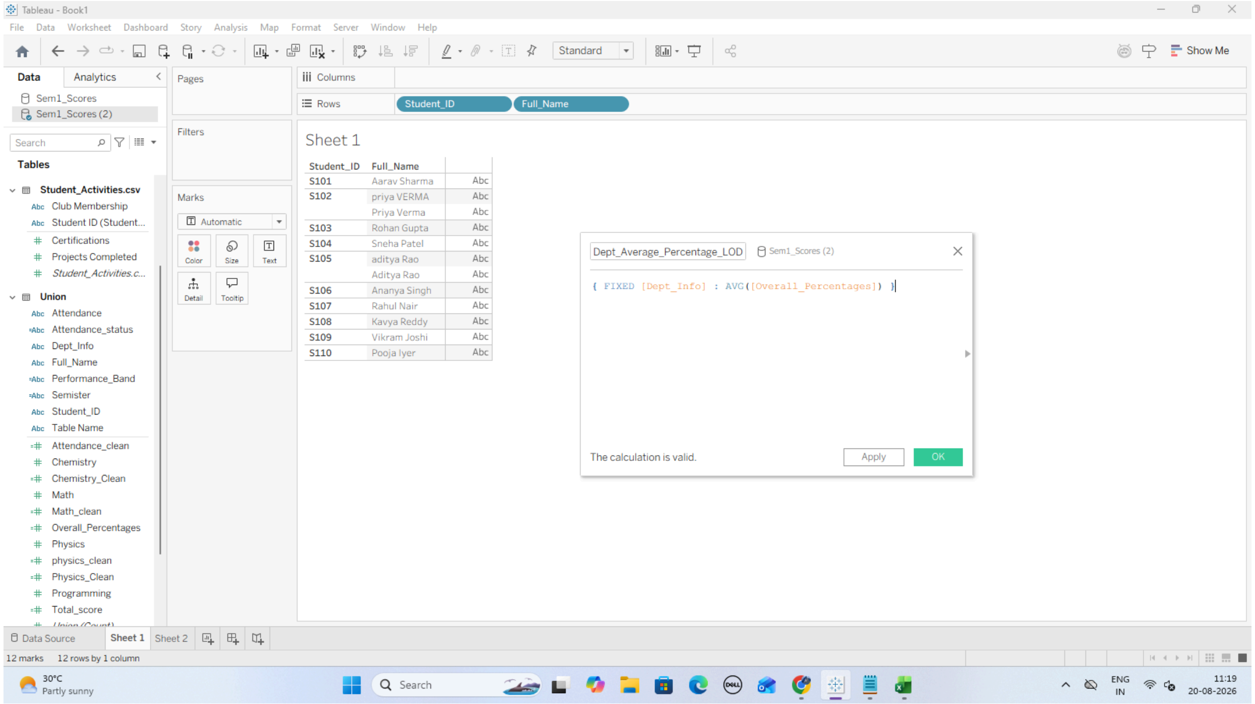
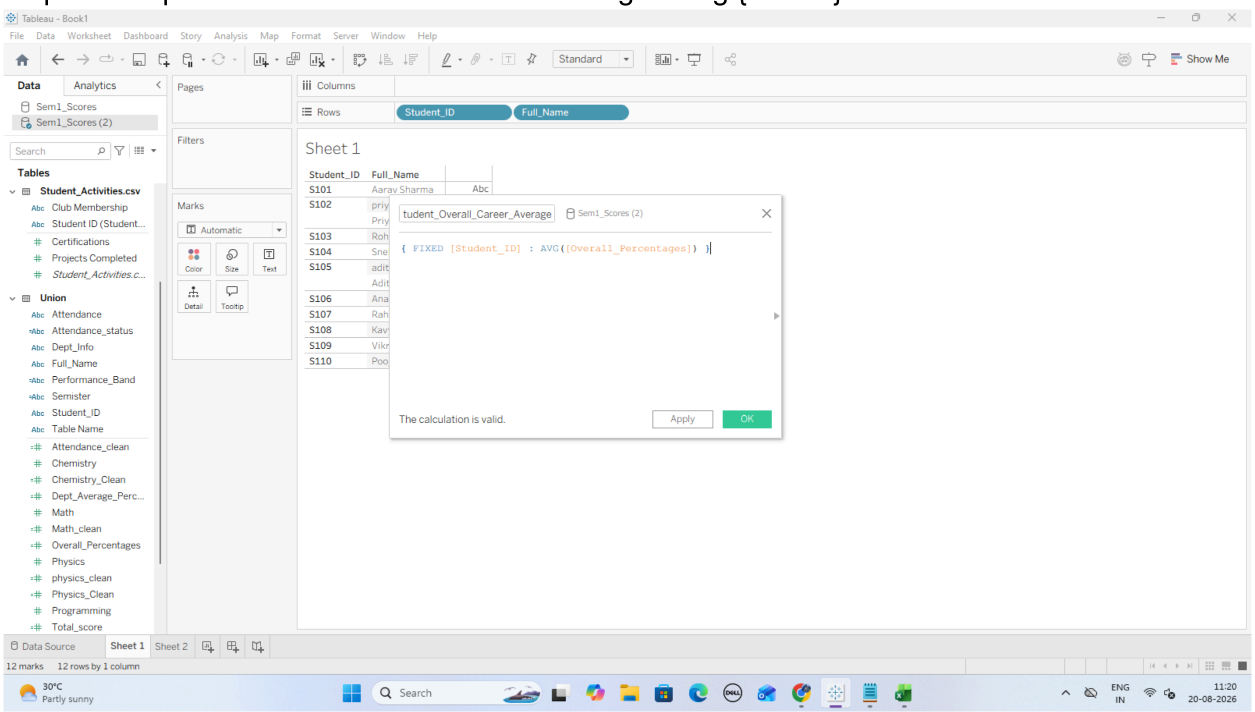
F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD



Step 2: Compute Student Multi-Semester Average using {FIXED} LOD



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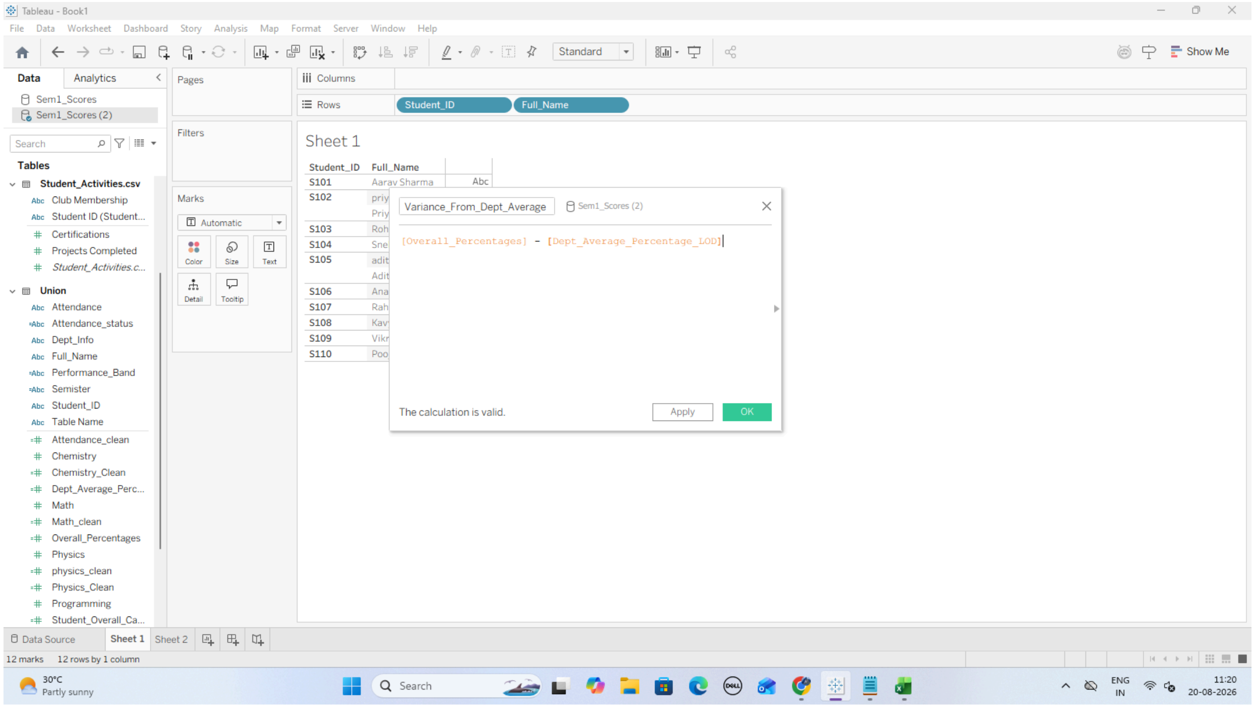
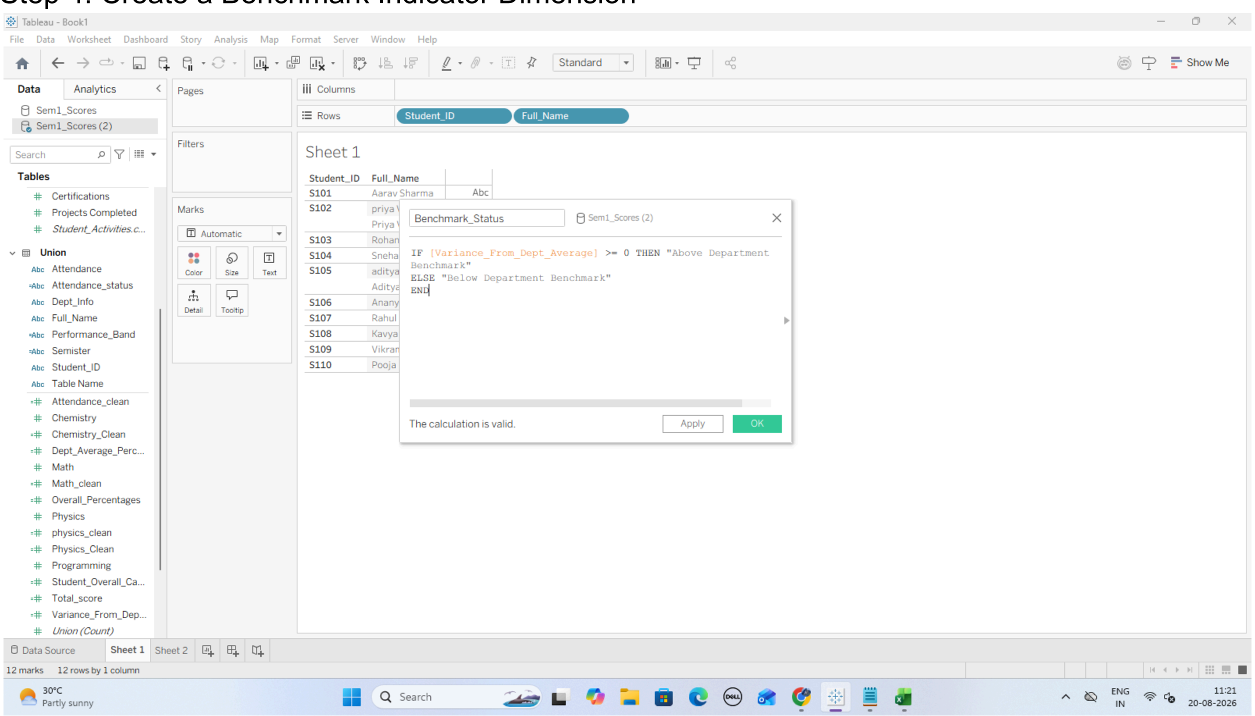
Step 3: Calculate Performance Variance (Student vs. Department Benchmark)



Step 4: Create a Benchmark Indicator Dimension



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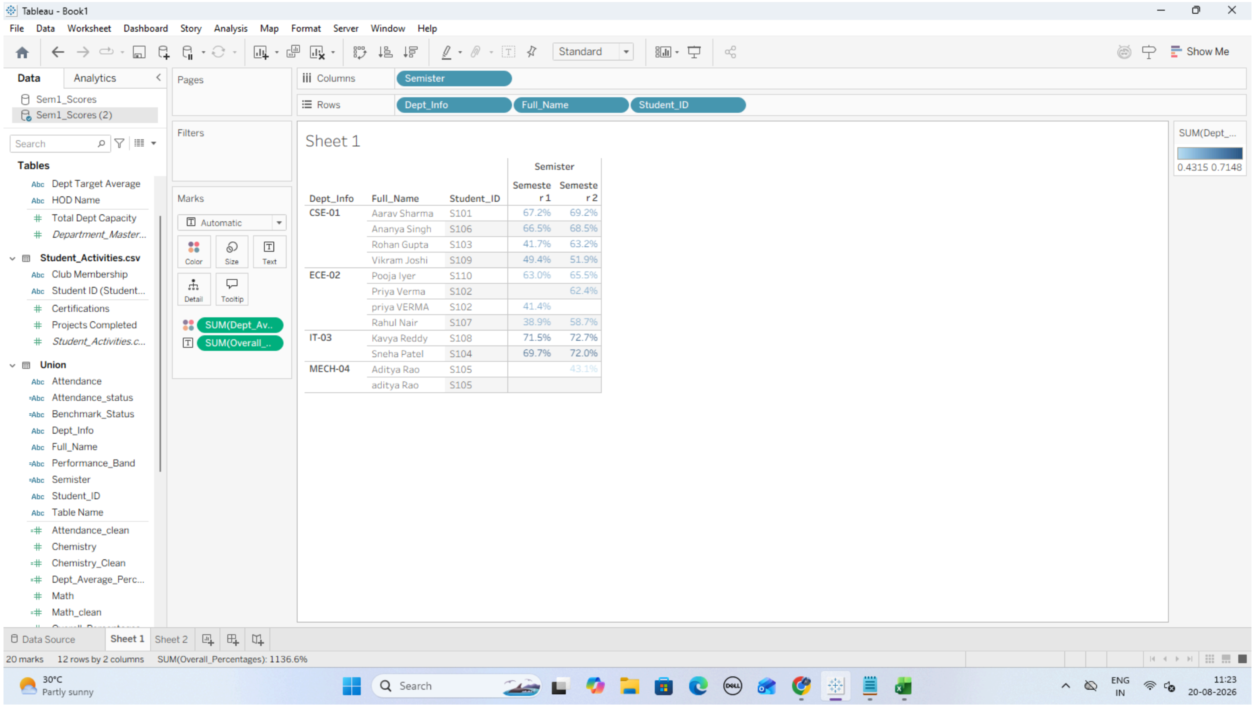
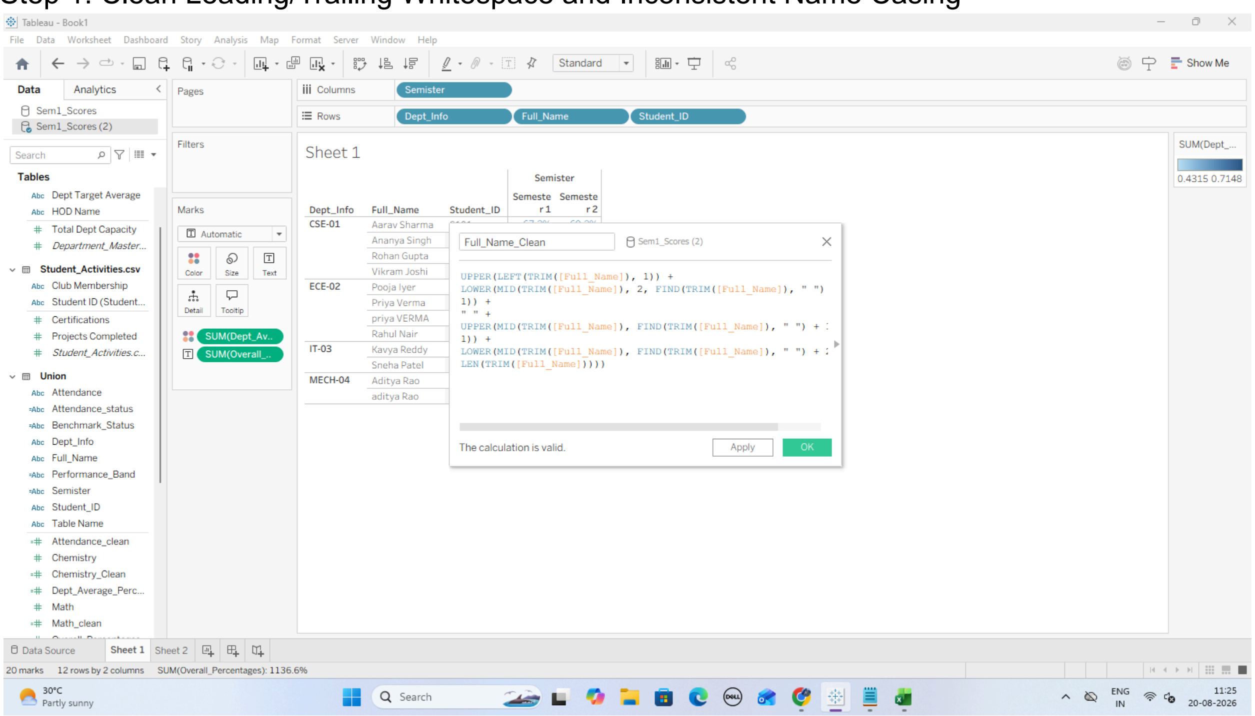
Step 5: Build the LOD Benchmark Visualization



G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing



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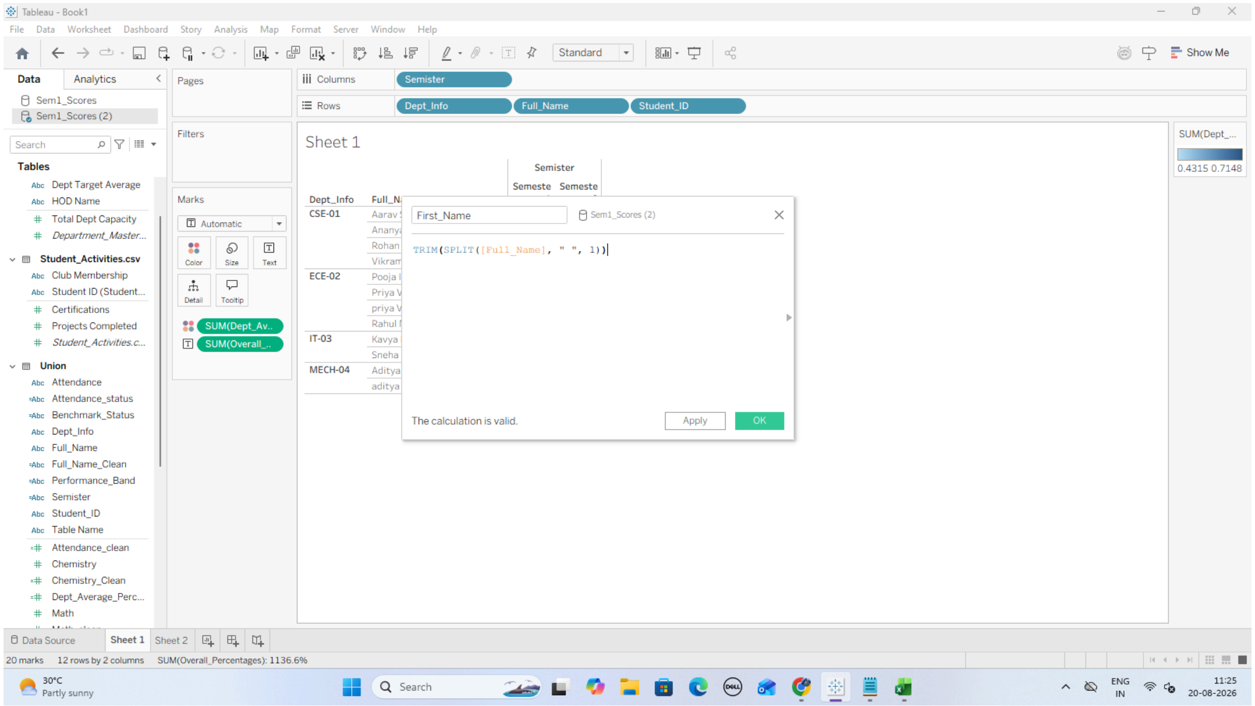
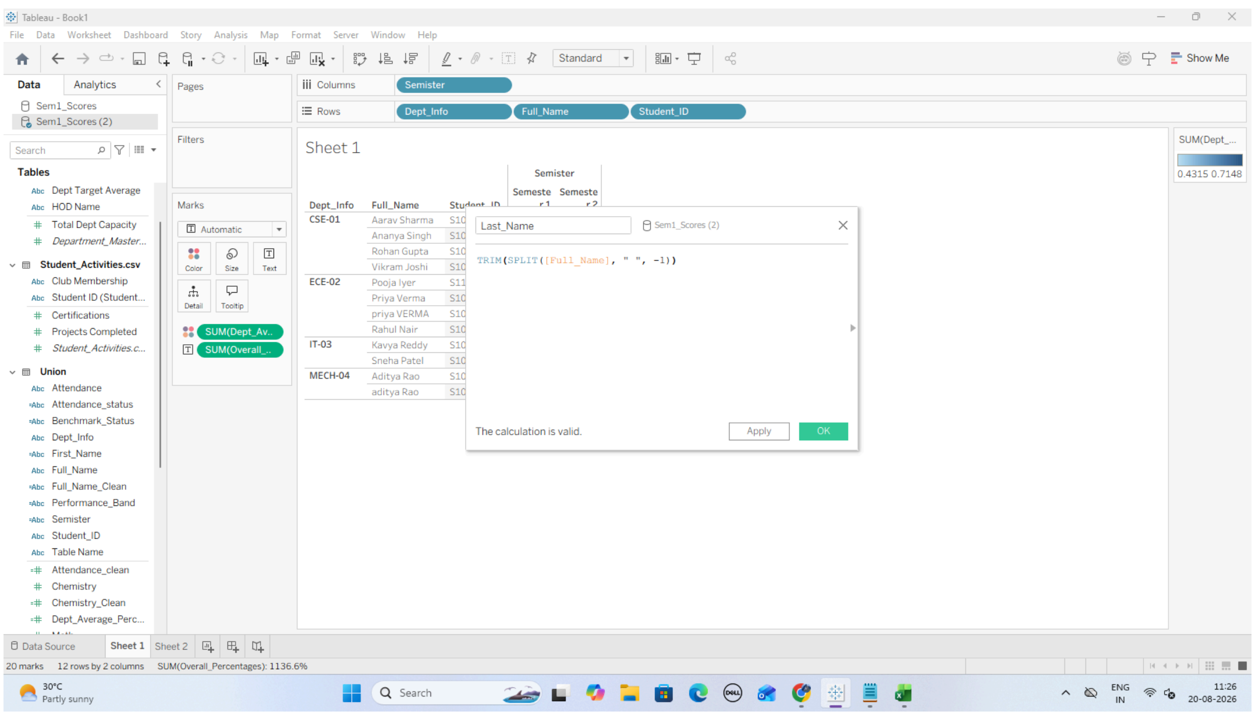
Step 2: Parse First Name and Last Name using SPLIT()





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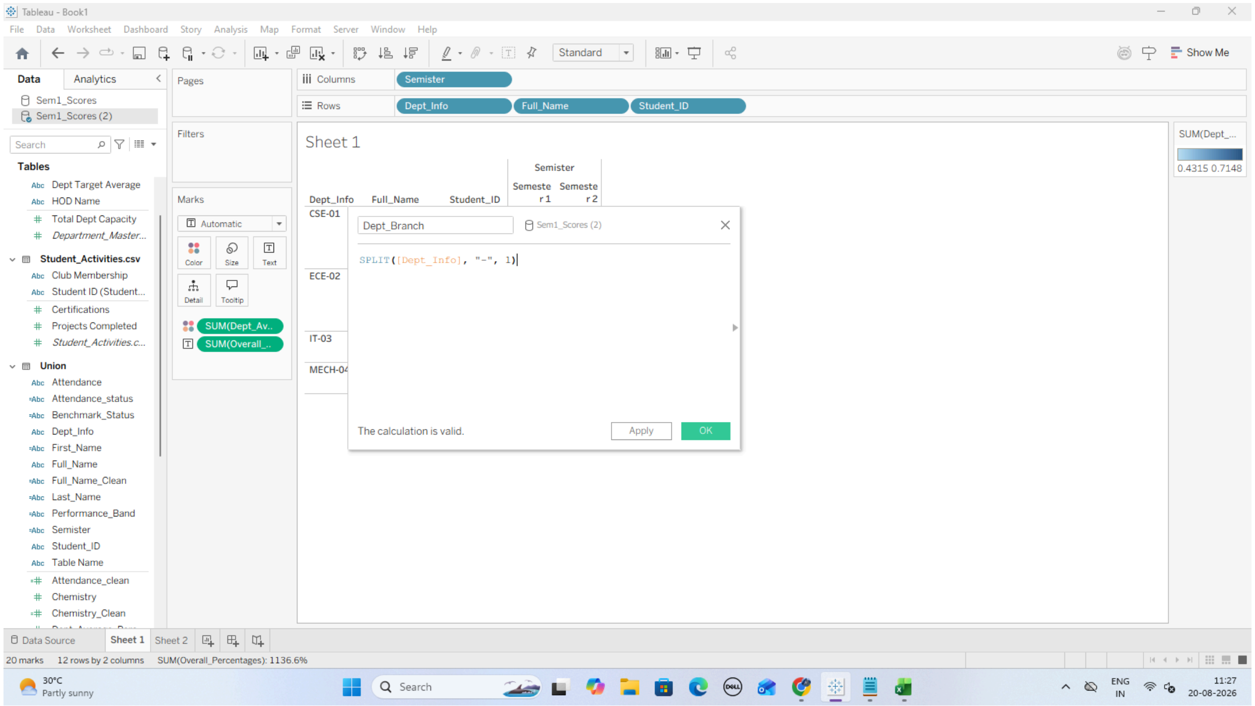
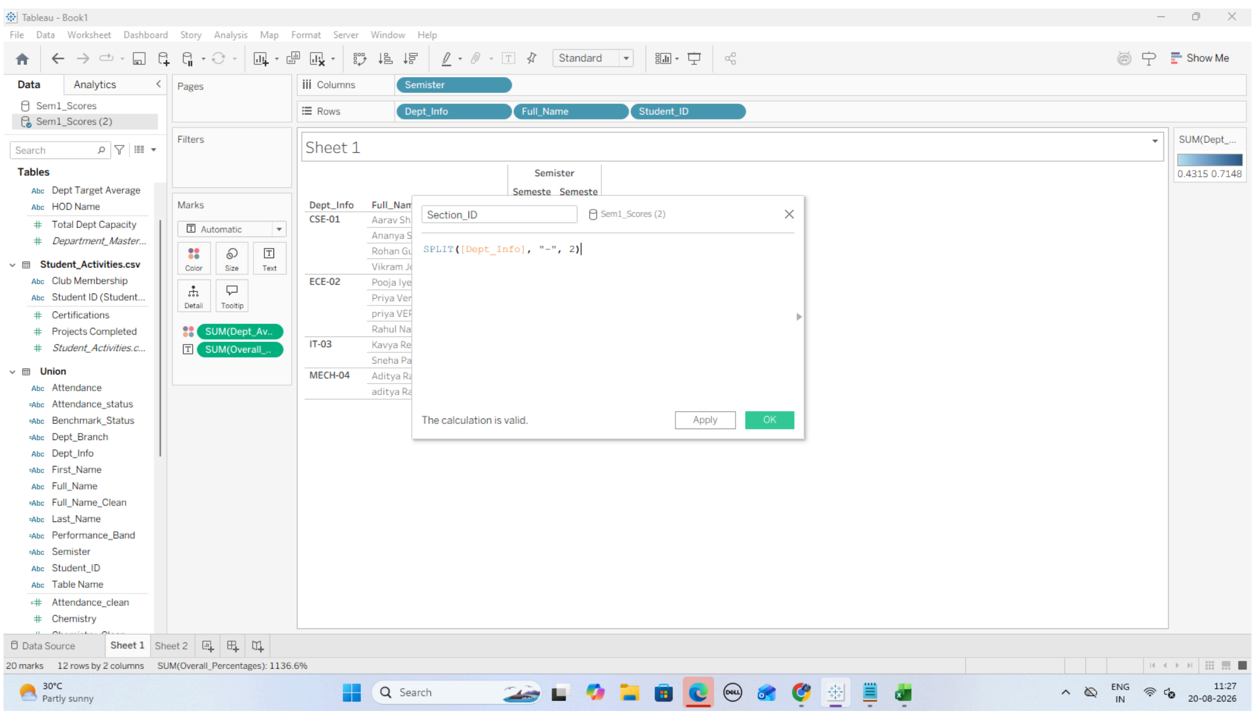
Step 3: Split Composite Department Codes (Dept_Info)





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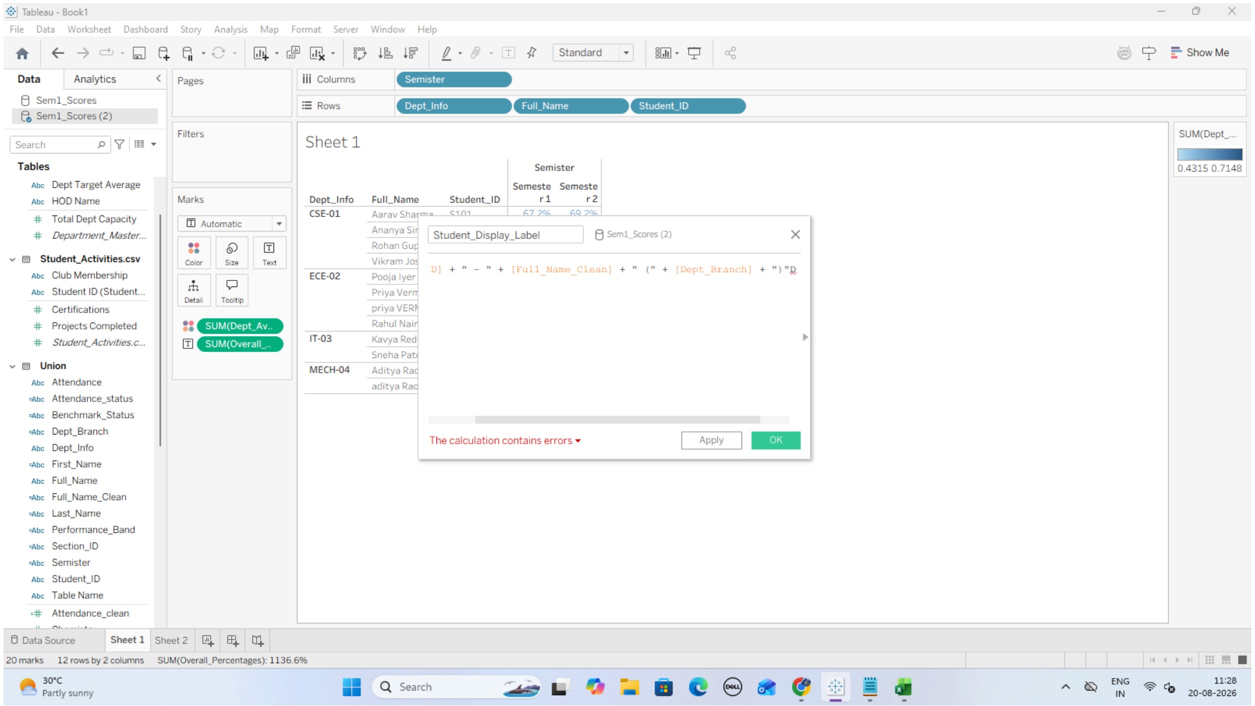
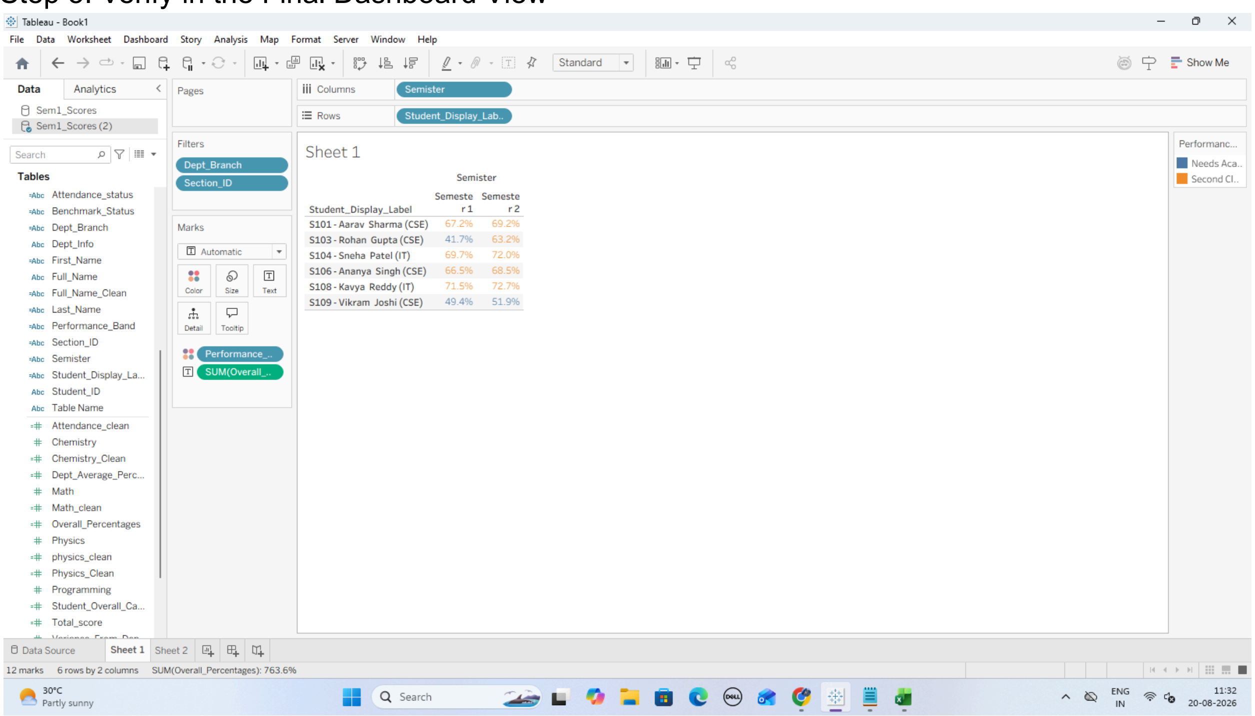
Step 4: Create a Clean Composite Student Label



Step 5: Verify in the Final Dashboard View



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